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B.Tech. Degree VI Semester Examination April 2018

ME 15-1605 CAD/CAM
(2015 Scheme)

Time : 3 Hours

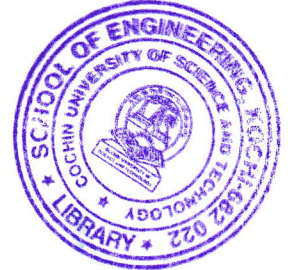
Maximum Marks : 60

PART A

(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) Explain any four geometric transformations used in CAD with appropriate transformation matrices.
- (b) What is computer aided process planning (CAPP)? What are the challenges in developing generative type CAPP?
- (c) List out and explain different levels of automation.
- (d) What is line balancing? What is balance delay?
- (e) Explain the terms (i) NC (ii) CNC (iii) DNC and (iv) FMC.
- (f) What are the different types of statements used in APT? Explain each with an example.
- (g) Explain the different static and dynamic tests performed on CNC machines.
- (h) Explain the types and features of spindle bearings for machine tools.
- (i) Explain the structure of a SCARA robot with the help of a neat figure.
- (j) Compare textual language programming with teaching method in robotics.



PART B

(4 × 10 = 40)

- II. (a) Explain (i) extrusion (ii) revolve (iii) loft and (iv) sweep functions in solid modeling with the help of neat figures. (8)
- (b) Explain constructive solid geometry with the help of an example. (2)

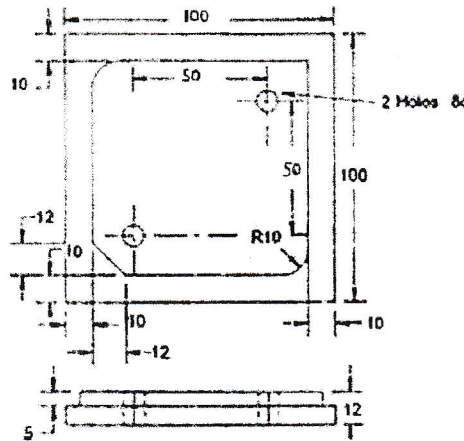
OR

- III. Draw the network diagram and solve the following assembly line balancing problem using Largest Candidate Rule. The production rate required is one assembly every 15 min. Find balance delay. (10)

Task	Time	Immediate Predecessors
A	12	Nil
B	6	A
C	6	B
D	2	B
E	2	B
F	12	B
G	7	C, D
H	5	G
I	1	E
J	4	I, F
K	6	H, J
L	7	K

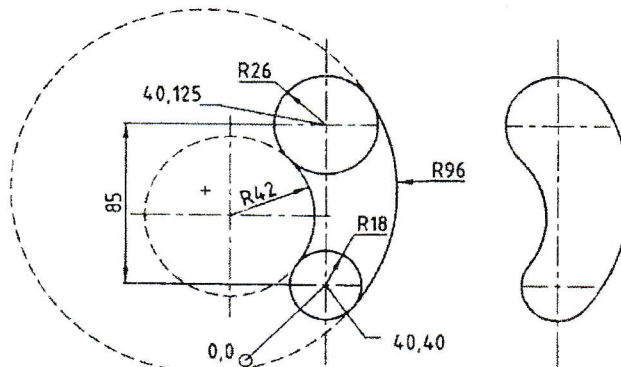
(P.T.O.)

- IV. A component to be machined is shown in the figure below. The outer profile (step 5 mm deep) needs to be machined using an end mill of diameter 16 mm (HSS). Prepare the NC code. (10)
(The work piece provided is 100 mm × 100 mm × 12 mm)



OR

- V. A component as shown in the figure below is to be machined from an MS sheet of 5 mm thickness. Write APT code for milling the profile with a $\phi 10$ mm end mill. (10)



- VI. Explain the working principles of an optical rotary encoder for CNC machines with the help of neat figures. A ball screw has a pitch of 10 mm. How many radial gratings are required for an accuracy of 1 μ m if two optical sensors are used at 90° phase difference? (10)
- OR
- VII. What are modular tools? List out and explain the components of modular tools with neat figures. Explain the cost advantage in using modular tools compared to custom made tools. (10)
- VIII. Explain the criteria in the selection of Robotic configurations programming techniques, payload and accuracy for their applications in (i) assembly (ii) arc welding. (10)
- OR
- IX. Explain with appropriate illustrations (i) hydraulic and pneumatic components, valves, actuators and their control (ii) AC and DC servo motors and their control (iii) various sensors used in electric, hydraulic and pneumatic systems, and sensors used in robotics such as vision systems, range detectors, force and torque sensors. (10)